Day 23 | Monday

23

December 7, 2026

Carpentry drawing: floor layout and cut list

12:45 PM–2:00 PM • Math • 75 min

Materials practice: sheeting and allowance

2:15 PM–3:30 PM • Calculation Practice • 75 min

Week 7 • 2 scheduled blocks • 150 teaching minutes

Estimate. Show the method. Label units. Check the result.

Today’s exact schedule

12:45 PM–2:00 PM • 75 min

Carpentry drawing: floor layout and cut list

12:45 PM–12:50 PM	**Start**	5
12:50 PM–12:55 PM	**Recall**	5
12:55 PM–1:10 PM	**Teach**	15
1:10 PM–1:25 PM	**Together**	15
1:25 PM–1:45 PM	**Practise**	20
1:45 PM–1:55 PM	**Check**	10
1:55 PM–2:00 PM	**Wrap up**	5

Math • Math Week 7, Lesson 1

2:15 PM–3:30 PM • 75 min

Materials practice: sheeting and allowance

2:15 PM–2:20 PM	**Start**	5
2:20 PM–2:25 PM	**Recall**	5
2:25 PM–2:40 PM	**Teach**	15
2:40 PM–2:55 PM	**Together**	15
2:55 PM–3:15 PM	**Practise**	20
3:15 PM–3:25 PM	**Check**	10
3:25 PM–3:30 PM	**Wrap up**	5

Calculation Practice • Math Week 7, Lesson 2

Break: 2:00 PM–2:15 PM • 15 minutes • excluded from teaching time

Carpentry drawing: floor layout and cut list

Today’s goal: Extract dimensions from a drawing, calculate a perimeter and offcut, and explain assumptions in a cut list.

Identify one target skill and the checking habit to use today.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Exercise Book 10B Q1 (p.95); 3C Q3 (p.29); 12B Q5 (p.117)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Recall and reconnect

Discuss which dimensions are needed for perimeter, area and diagonal; retrieve one kerf calculation.

Review previous feedback and attempt the retrieval prompt without displayed solutions.

Resources: Exercise Book 10B Q1 (p.95); 3C Q3 (p.29); 12B Q5 (p.117)

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Match each floor measurement to its purpose

1 The floor is a 7.2 m × 3.6 m rectangle: label both dimensions before choosing a calculation.

2 Perimeter measures the boundary: 2(7.2 + 3.6) = 21.6 m. Area measures coverage: 7.2 × 3.6 = 25.92 m².

3 The diagonal crosses the rectangle: √(7.2² + 3.6²) ≈ 8.05 m. It is a length, not an area.

4 Keep the rectangular layout and no-waste assumptions visible. This is the instructor's plywood adaptation of 10B Q1.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Keep the floor and stock calculations separate

1. Exercise Book 3C Q3 is a separate 6,000 mm stock problem; its dimensions do not come from the floor drawing.

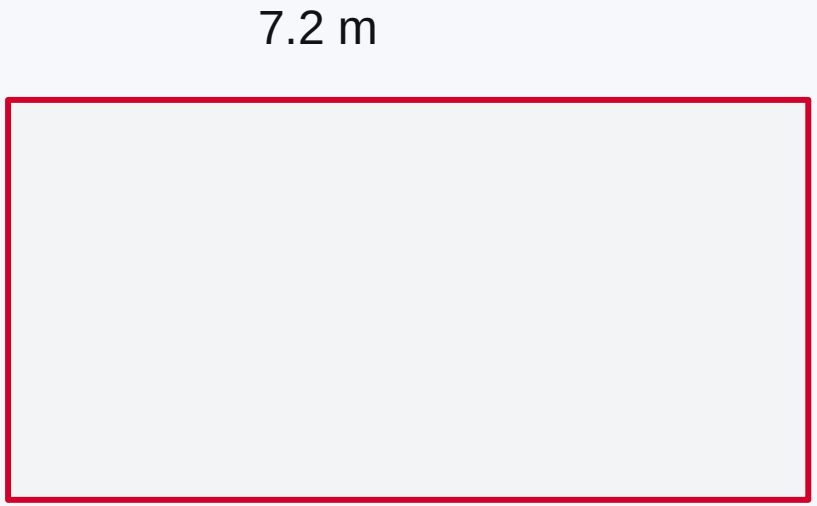
2. Keep all four stated cuts and the 3 mm kerf in the stock equation: stock − total pieces − total kerf = offcut.

3. Check backwards: total pieces + total kerf + offcut must equal the original stock length.

4. Use metres for the floor and millimetres for the stock. Attach each unit and assumption to the correct sketch.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

One floor, three different measurements



Boundary: $P = 2(7.2 + 3.6)$
$= 21.6$ m

Coverage: $A = 7.2 \times 3.6$
$= 25.92$ m²

Diagonal: $\sqrt{(7.2^2 + 3.6^2)} \approx 8.05$ m.
Keep rectangular-layout and no-waste assumptions explicit.

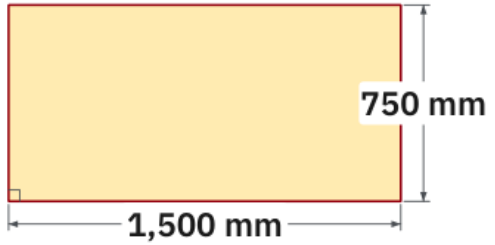
Together • Annotate this floor and a separate 6,000 mm stock sketch.
Identify which information belongs in each calculation before using tools.

Instructor carpentry adaptation of 10B Q1; plywood replaces checker plate • Shape Canvas units: m • Schematic

Textbook | page 220

Example 9.1 A rectangular guard frame

A guard frame is 1,500 mm long and 750 mm wide. How much flat bar goes around it?



SOLUTION

1. Two lengths and two widths: P = 2 × (l + w).
2. P = 2 × (1,500 + 750) = 2 × 2,250 = 4,500

Answer **4,500 mm, which is 4.5 m.**

Example 9.1: label the boundary before calculating a perimeter.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Spot the mistake. Explain the correction.

A cut-list equation subtracts the required pieces from 6,000 mm but has no kerf term. What quantity is missing?

Discuss the method before checking the answer.

Include the material lost to all four stated cuts at 3 mm per cut. Keep this stock calculation separate from the floor, then check that pieces, kerf and offcut total 6,000 mm.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Work together

Partners annotate the floor rectangle and a separate 6,000 mm stock sketch from Exercise Book 3C Q3. Decide which information belongs to each calculation.

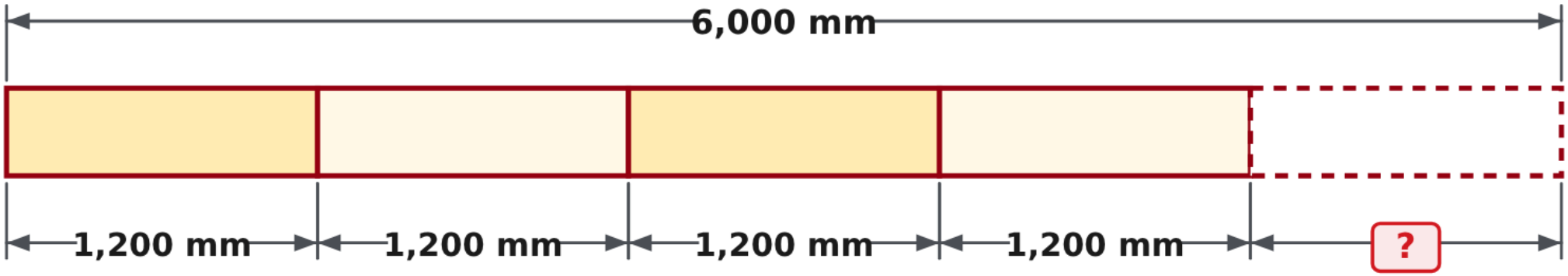
Explain the method and check directions; work with a partner only where permitted.

Resources: Exercise Book 10B Q1 (p.95); 3C Q3 (p.29); 12B Q5 (p.117)

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Guided practice | 3C Q3

Four 1,200 mm pieces are cut from a 6,000 mm bar with four cuts. The kerf is 3 mm. The offcut is 6,000 − (4 × 1,200 + 4 × 3). Find it.

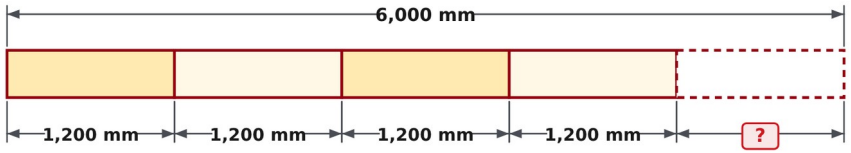


Work with a partner. Take turns explaining the next step.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Compare your method | 3C Q3

Four 1,200 mm pieces are cut from a 6,000 mm bar with four cuts. The kerf is 3 mm. The offcut is 6,000 − (4 × 1,200 + 4 × 3). Find it.



CHECK TOGETHER

1,188 mm

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Your scheduled practice

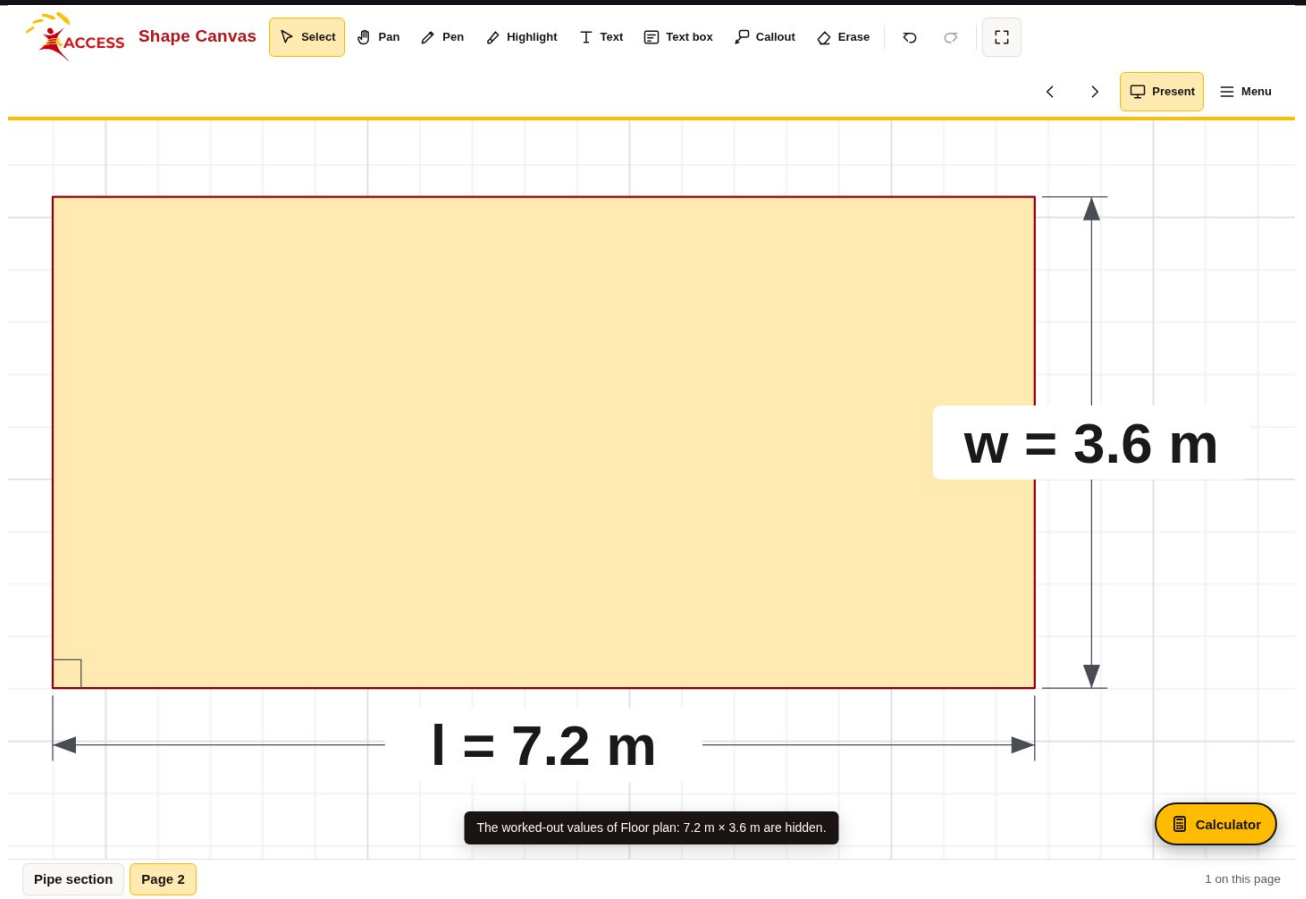
Draw and label the floor in Shape Canvas. Calculate perimeter, floor area and diagonal; solve Exercise Book 3C Q3 without changing its four cuts or 3 mm kerf. Record units and assumptions; do not mix the two problem contexts.

Keep numbered written solutions and units. Use tools only under the conditions stated for this block.

Resources: Exercise Book 10B Q1 (p.95); 3C Q3 (p.29); 12B Q5 (p.117)

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Open Shape Canvas



Navigation example. Use the values and units in the assigned task.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Practise in Numeracy Hub and Shape Canvas

1. In Shape Canvas, set units to m and draw the 7.2 m × 3.6 m floor rectangle; label the dimensions.

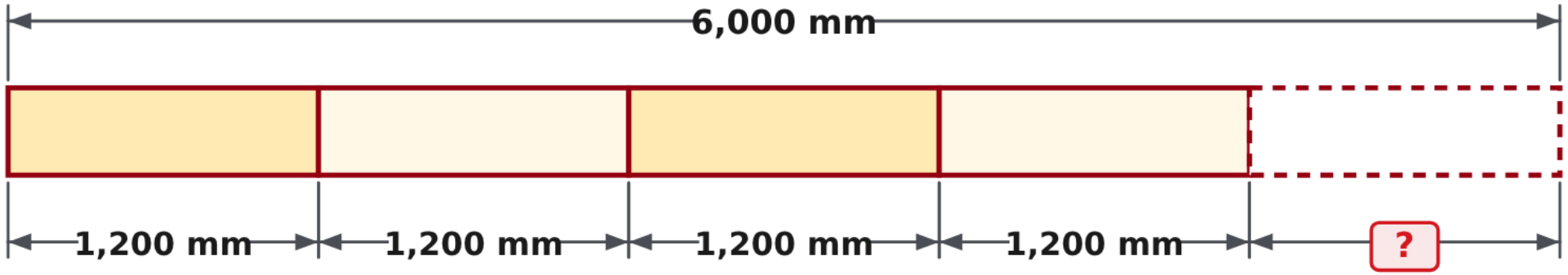
2. Add a Text cut-list note that keeps the separate 6,000 mm stock problem, four cuts and 3 mm kerf explicit; retain the written equations and assumptions.

3. After the paper calculations, use Numeracy Hub Calculator for numerical checks. Save the labelled floor sketch and assumptions for the next material-order block.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Practise from the Exercise Book | 3C Q3

Four 1,200 mm pieces are cut from a 6,000 mm bar with four cuts. The kerf is 3 mm. The offcut is 6,000 − (4 × 1,200 + 4 × 3). Find it.



Write the operation, working and units before checking.

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Check independently

Submit the labelled floor plan, perimeter calculation and stock offcut equation.

Submit work and demonstrate the checking strategy; identify a remaining difficulty.

Resources: Exercise Book 10B Q1 (p.95); 3C Q3 (p.29); 12B Q5 (p.117)

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Wrap up and choose the next step

Identify what improved, what needs another attempt and which resource will support the next step.

Name the math and another essential skill used; record a concrete follow-up action.

Resources: Exercise Book 10B Q1 (p.95); 3C Q3 (p.29); 12B Q5 (p.117)

Textbook Chapter 3 pp.68–75; Chapter 9 pp.220–221; Chapter 10 pp.242–245

Break

2:00 PM–2:15 PM

Return ready for: Materials practice: sheeting and allowance

This break is excluded from teaching minutes.

Materials practice: sheeting and allowance

Today’s goal: Use area and a stated classroom allowance to build a material quantity and distinguish an estimate from a cutting layout.

Identify one target skill and the checking habit to use today.

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Exercise Book 10B Q1–2 (p.95); 6B Q2 (p.57)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Recall and reconnect

Retrieve the floor area and one-sheet area from the previous block. Explain why a fractional purchase quantity rounds up.

Review previous feedback and attempt the retrieval prompt without displayed solutions.

Resources: Exercise Book 10B Q1–2 (p.95); 6B Q2 (p.57)

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

An area count needs a matching layout

1 The floor area is 25.92 m². One 1.2 m × 2.4 m sheet covers 2.88 m².

2 The no-waste area calculation gives 25.92 ÷ 2.88 = 9 sheets.

3 Three 2.4 m lengths make 7.2 m; three 1.2 m widths make 3.6 m. This 3 × 3 arrangement verifies the nine-sheet rectangular layout.

4 An area estimate alone does not prove that every cutting layout will fit. Here, both dimensions are checked against the arrangement.

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Separate the base quantity from the allowance

1 Keep the nine-sheet no-waste base visible before adding any allowance.

2 For this classroom exercise only, the stated 10% allowance gives 9 × 1.10 = 9.9 sheets.

3 An order needs whole sheets, so round 9.9 up to 10. The additional sheet belongs to the allowance calculation.

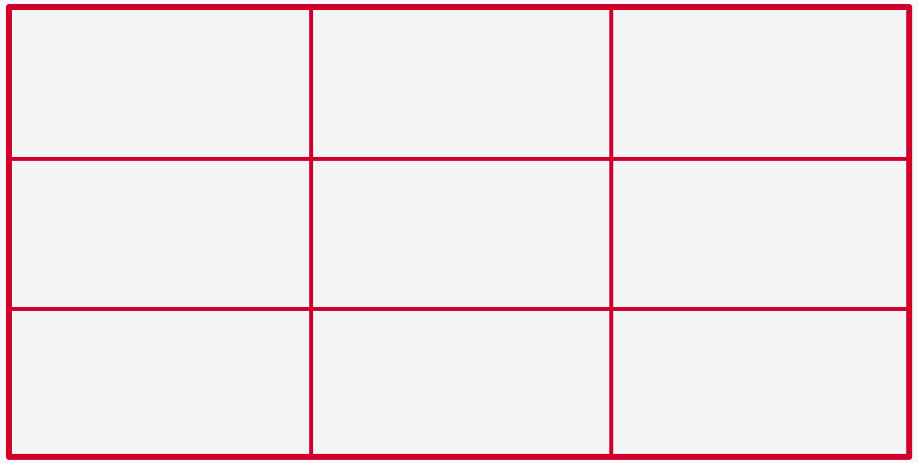
4 The 10% is instructor-provided problem data, not a universal site allowance. Do not add prices or other unstated job assumptions.



Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Verify the base layout before adding allowance

3 × 2.4 m = 7.2 m



3 × 1.2 m = 3.6 m • 9 sheets

Base: 25.92 ÷ 2.88 = 9
With 10%: 9 × 1.10 = 9.9
Whole-sheet order: 10

The nine-sheet arrangement verifies this simple rectangular base layout. The tenth sheet is the stated classroom planning allowance.

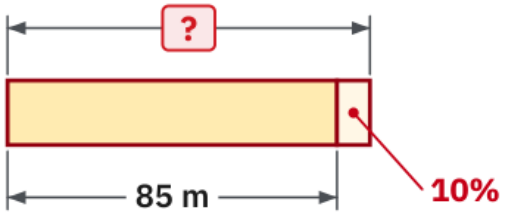
Together • Recreate the 3 × 3 layout in Shape Canvas. Compare the exact no-waste layout with the separate allowance quantity; explain final rounding.

Instructor adaptation • Sheet size 1.2 m × 2.4 m • 10% is classroom data, not a universal job allowance

Textbook | page 142

Example 6.5 Ordering pipe with a waste allowance

A job needs 85 m of pipe. Add a 10% waste allowance. The pipe is sold in 6 m lengths. How many lengths do you order?



SOLUTION

1. 10% of 85 = 0.1 × 85 = 8.5 m.
2. Add it on: 85 + 8.5 = 93.5 m.
3. Lengths: 93.5 ÷ 6 = 15.5833...
4. A part length still has to be bought, so round up to 16.

Example 6.5: a stated allowance is added to the original quantity.

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Textbook | page 142

CALCULATOR

TI-30Xa

85 | + | 10 | 2nd | % | =

Display: 93.5

Sharp EL-531XT

85 | + | 10 | 2ndF | %

Display: 93.5

Pressing + and then % adds 10% of 85 to 85 on both calculators (TI-30Xa guidebook, p. 4, and Sharp EL-531XT Calculation Examples [4]). In one step without the % key: 85 × 1.1 = 93.5.

Answer **93.5 m of pipe, which is 16 lengths.**

Example 6.5: compare calculator entries and the whole-length order.

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Spot the mistake. Explain the correction.

A quantity note shows 9.9 sheets and orders 9 because the base layout used nine. Why is the order short?

Discuss the method before checking the answer.

The verified base and the allowance order answer different questions. Keep the base at nine, retain the stated 10% calculation, and round the final purchase quantity up to ten whole sheets.

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Work together

Pairs lay out nine sheets as three 2.4 m lengths by three 1.2 m widths. Compare the verified no-waste layout with the separate allowance quantity.

Explain the method and check directions; work with a partner only where permitted.

Resources: Exercise Book 10B Q1–2 (p.95); 6B Q2 (p.57)

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Your scheduled practice

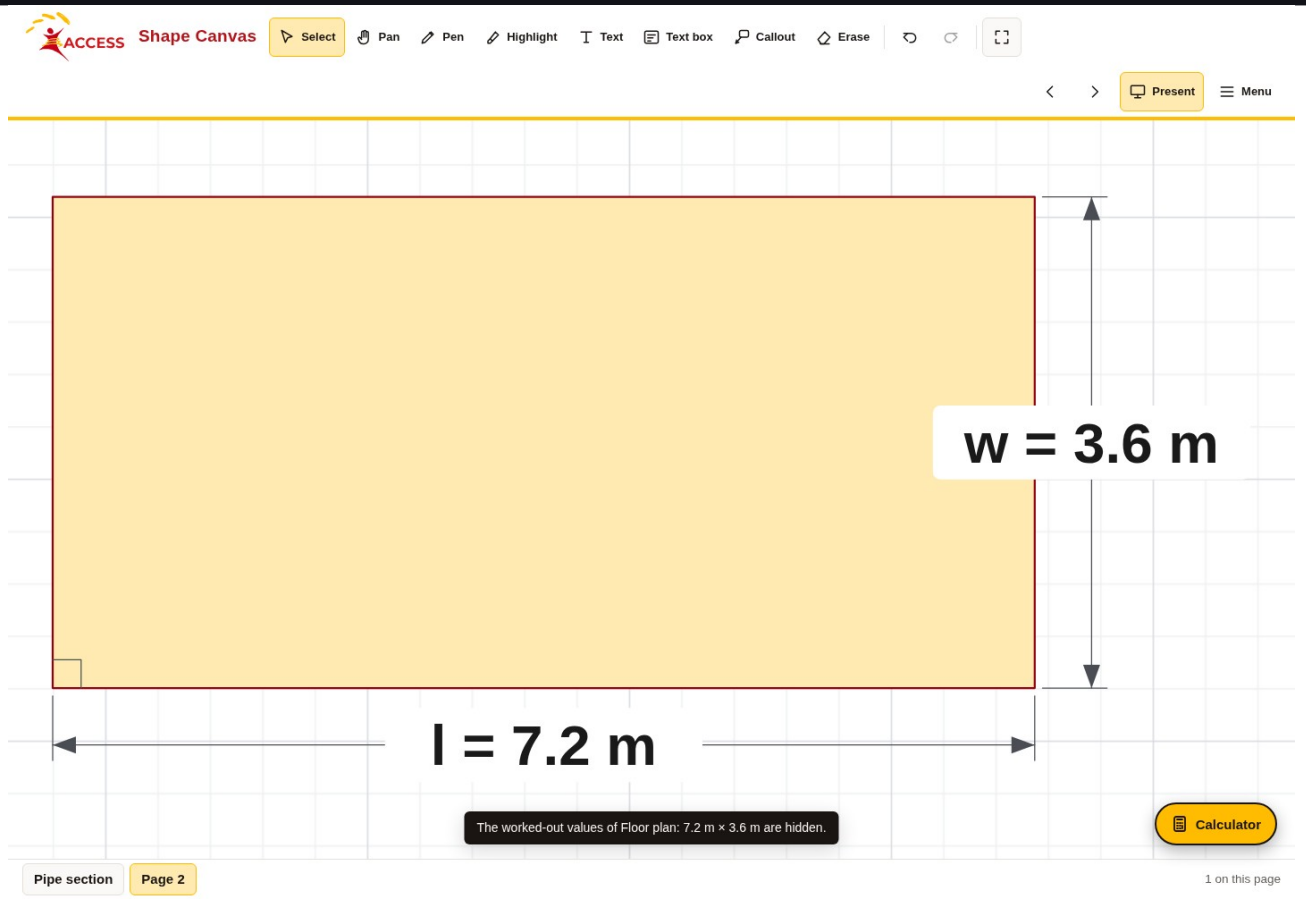
Prepare a quantity note: floor area, sheet area, nine-sheet base and stated 10% allowance. Then solve Exercise Book 10B Q2 for coating tins. Check units, final rounding and assumptions; no prices are assumed.

Keep numbered written solutions and units. Use tools only under the conditions stated for this block.

Resources: Exercise Book 10B Q1–2 (p.95); 6B Q2 (p.57)

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Open Shape Canvas



Navigation example. Use the values and units in the assigned task.

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Practise in Numeracy Hub and Shape Canvas

1. Open the floor sketch in Shape Canvas and use 1.2 m × 2.4 m sheet rectangles to recreate the scheduled 3 × 3 arrangement.

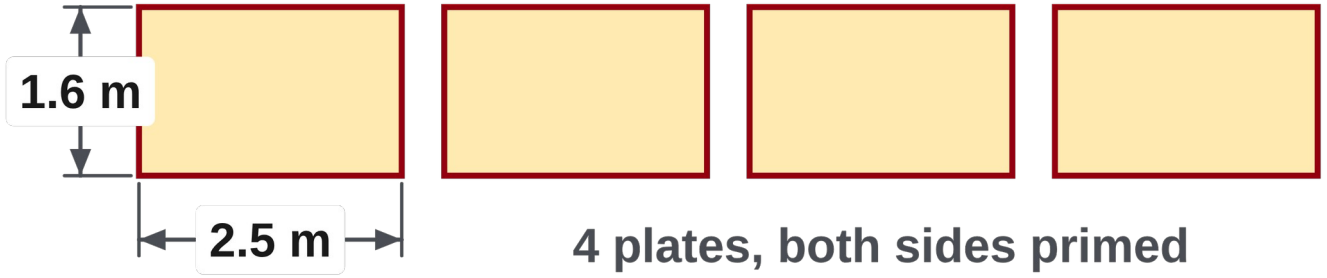
2. Check the arrangement against the 7.2 m and 3.6 m floor dimensions; record the floor area, sheet area and nine-sheet base in the written quantity note.

3. Use Numeracy Hub Calculator after paper working to compare the no-waste count with 9 × 1.10. Record the stated allowance and final whole-sheet rounding separately.

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Practise from the Exercise Book | 10B Q2

Four plates, each 2.5 m by 1.6 m, are primed on both sides. The primer label says 1 L covers 7 m². Primer is sold in 1 L tins. How many tins do you buy? Give the litres to two decimal places.



4 plates, both sides primed

1 L covers 7 m²

Litres: ?

1 L tins: ?

Write the operation, working and units before checking.

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Check independently

Explain the nine-sheet base, allowance order, and why an area estimate alone does not prove every cutting layout.

Submit work and demonstrate the checking strategy; identify a remaining difficulty.

Resources: Exercise Book 10B Q1–2 (p.95); 6B Q2 (p.57)

Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254

Wrap up and choose the next step

Identify what improved, what needs another attempt and which resource will support the next step.

Name the math and another essential skill used; record a concrete follow-up action.

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Textbook sections 6.4 pp.142–143; Chapter 10 pp.242–254